

Linear Algebra—Winter/Spring 2026

Instructors: A.S. Arora, S. Das, G. R. Kurri

Problem Set 11

(Last Edited: Friday, April 17, 2026)

Due: 10 PM, Wednesday April 22, 2026

Instructions

- a. *Timeline.* In general, expect the first “half” of a problem set to be released by Tuesday nights (or whenever the first lecture of the week takes place). Expect the full problem set to be released by Friday nights (or whenever the second lecture of the week takes place). The full assignment will, *typically*, be due on Tuesday nights, 8 PM.
- b. *Punctuality.* Aim to get the assignment done three days prior to the deadline. We will not be able to entertain any requests for extensions. If you seek clarification from us and we do not respond on time, do the following: (a) assume whatever seems most reasonable, and solve the question under that assumption and (b) ensure you have evidence for making the clarification request and report the incident to one of the PhD TAs.
- c. *Resources.* Feel free to use any resource you like (including generative AI tools) to get help but please make sure you understand what you finally write and submit. The TAs may ask you to explain the reasoning behind your response if something appears suspicious.
- d. *Graded vs Practice Questions.* Problems numbered using Arabic numerals (i.e. 1, 2, 3, ...) constitute assignment problems that will be graded. Problems numbered using Roman numerals (i.e. i, ii, iii, ...) are practice problems that will not be graded.

The text here is Linear Algebra Done Right, by Sheldon Axler (3rd edition); we are on Chapter 7.

There will be no further submission deadline extension as we have already relaxed it by one day.

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- i. Let $T, T' \in \mathcal{L}(V, W)$. Prove that if for all $w \in W$, it holds that $\langle w, Tv \rangle = \langle w, T'v \rangle$ then $Tv = T'v$. Also show that if $Tv = T'v$ for all $v \in V$, then $T = T'$.
 1. Fix $u \in V$ and $w' \in W$. Define $T \in \mathcal{L}(W, V)$ by

$$Tw = \langle w, w' \rangle u$$

for every $w \in W$. Find a formula for T^* .

2. Understand and write down the proof of *Theorem 7.36* in the textbook, Section 7C that states *Every positive operator on V has a unique positive square root.*
3. These are used as intermediate steps in the proof of Proposition 7.42 (Characterisation of Isometries). Prove them (you can assume these to be true when writing the proof of Proposition 7.42).

a. Suppose V is a real inner product space. Prove that

$$4 \cdot \langle u, v \rangle = \|u + v\|^2 - \|u - v\|^2.$$

b. Suppose V is a complex inner product space. Prove that

$$4 \cdot \langle u, v \rangle = \|u + v\|^2 - \|u - v\|^2 + \|u + \iota v\|^2 - \|u - \iota v\|^2$$

where ι is the square root of -1 .

c. Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST = I$ iff $TS = I$.

4. Understand and write down the proof of *Proposition 7.42 Characterisation of Isometries* in the textbook, Section 7C that states

Suppose $S \in \mathcal{L}(V)$. Then the following statements are equivalent:

- a. *S is an isometry;*
- b. *$\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$;*
- c. *$Se_1, \dots, Se_n$ is orthonormal for every orthonormal list of vectors $e_1, \dots, e_n$ in V ;*
- d. *there exists an orthonormal basis $e_1, \dots, e_n$ of V such that $Se_1, \dots, Se_n$ is orthonormal;*
- e. *$S^*S = I$;*
- f. *$SS^* = I$;*
- g. *S^* is an isometry*
- h. *S is invertible and $S^{-1} = S^*$.*

- ii. Consider the following statement.

Suppose V is a real inner product space and $T \in \mathcal{L}(V)$. If $\langle Tv, v \rangle = 0$ for all $v \in V$, then $T = 0$.

- a. Is this statement true? If it is true, give a proof. Otherwise, give a counter-example.
- b. Suppose the field in the statement were complex. What can you say about the statement now? If it is still false, give a counter-example. Otherwise, give a proof.

- iii. Consider the linear operator $T(x, y) = (-y, x)$ on $T \in \mathcal{L}(R^2)$.

a. Show that $\langle Tv, v \rangle = 0$ for all $v \in R^2$.

- b. Show that if the field was complex, there exists a non-zero vector for which $\langle Tv, v \rangle \neq 0$.
- iv. Let $T \in \mathcal{L}(V)$ be a linear operator on a complex inner product space V . Write $\langle Tu, w \rangle$ as a sum of terms of the form $\langle Tv, v \rangle$ for some vector $v \in V$, i.e.

$$\langle Tu, w \rangle = \sum_{v \in V'} c_v \langle Tv, v \rangle$$

where $V' \subseteq V$ is a finite subset and c_v (for $v \in V'$) are scalars that do not depend on T .

- v. Show that $(ST)^* = T^*S^*$ for all $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, U)$.
- vi. Show that $(T^*T - TT^*)$ is self-adjoint, where T is a linear operator. Show that if T is normal, then $T - \lambda I$ is also normal.
- vii. Show that the null space of T is equal to that of T^* , for every normal operator T .
- viii. Suppose $T \in \mathcal{L}(V)$ is self-adjoint (i.e. $T^* = T$) and all the eigenvalues of T are nonnegative. Show that T has a positive square root.
(Hint. This is proved in the textbook as well, but they skip one step. Be sure to complete the proof.)
- ix. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$ as an operator $A \in \mathcal{L}(\mathbb{C}^2)$. Is A positive? Prove your answer.
- x. Consider the move-by-four linear operator $T \in \mathcal{L}(\mathbb{C}^6)$ defined as follows:

$$T(a_1, a_2, a_4, a_5, a_6) := (a_5, a_6, 0, 0, 0, 0).$$

Find a square root of T , i.e. find an operator $R \in \mathcal{L}(\mathbb{C}^6)$ such that $R^2 = T$.

5. [Asks you to prove one step from the proof of Theorem 7.24 (Complex Spectral Theorem)]

Let V be a vector space over $\mathbb{C}$ and suppose $T \in \mathcal{L}(V)$ is a *normal* linear operator. Using Schur's Theorem, we know that the matrix of T is upper triangular on some orthonormal basis $e_1, \dots, e_n$, i.e.

$$\mathcal{M}(T, (e_1, \dots, e_n)) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ 0 & a_{2,2} & a_{2,3} & \dots & a_{2,n} \\ 0 & 0 & a_{3,3} & \ddots & \\ 0 & 0 & 0 & \dots & a_{n,n} \end{bmatrix}.$$

Suppose $a_{1,2} = 0$. Prove that $a_{2,3} = \dots = a_{2,n} = 0$.

- xi. Let σ be a permutation of $1, 2, \dots, n$. Define

$$\text{sign}(\sigma) = (-1)^{N_\sigma},$$

where N_σ is the number of pairwise exchanges (transpositions) required to transform σ into the identity permutation $(1, 2, \dots, n)$.

Prove that

$$\text{sign}(\sigma) = \text{sign}(\sigma^{-1}),$$

where σ^{-1} denote the inverse permutation of σ , i.e., the unique permutation satisfying $\sigma^{-1}(\sigma(i)) = i$ for all $i \in 1, 2, \dots, n$.